

Predicting Neural Network Architecture Viability After One Training Epoch via an Identity Deficit Threshold

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Abstract

We present a method for predicting whether a neural network architecture will form coherent task identity within a practical training budget using a single observation after one training epoch. The method introduces an identity deficit metric, $I_{\text{deficit}} = 1 - \sqrt{F}$, where F is the diagonal strength of the confusion matrix computed from epoch-one predictions. We find that architectures separate into two regimes at a critical threshold of $I_c = 0.22$: those that form coherent task identity and reach task competence, and those that do not within the tested training budgets. Validated across 22 architectures spanning multilayer perceptrons and convolutional networks on MNIST, Fashion-MNIST, and CIFAR-10, the method correctly classified 21 of 22 architectures (95%) using a consistent competence cutoff of 80% final accuracy across all three datasets. The single misclassification is a slow learner whose epoch-one identity deficit falls in a caution zone ($0.22 \leq I_{\text{deficit}} \lesssim 0.40$) that the method explicitly treats as requiring additional evaluation rather than immediate termination. The threshold was held fixed across the reported validation experiments. The method requires no access to gradients, weights, or loss curves, and produces a binary formation prediction from one confusion matrix in the time required to train a single epoch.

1 Introduction

Neural architecture search and hyperparameter optimization require practitioners to evaluate large numbers of candidate architectures. The standard workflow trains each candidate to near-convergence before its viability is known, meaning a search over one hundred architectures at fifty epochs per candidate requires one hundred full training runs, totaling five thousand training epochs, before the set of viable architectures is identified. A substantial fraction of those runs terminate in failure: the architecture never reaches useful performance on the target task regardless of how long training continues.

Existing approaches to this problem fall into three categories, each with a structural limitation that motivates the present work.

Early stopping methods monitor validation loss or accuracy over training and halt when improvement plateaus [4]. Methods such as Successive Halving and Hyperband extend this idea by allocating more resources to promising configurations across multiple stages [5]. These approaches are reactive: they require enough training epochs to observe a trend before any decision is possible, and they

optimize for training efficiency of a single run rather than discriminating viable from non-viable architectures as early as possible.

Learning curve extrapolation methods fit parametric curves to early training trajectories and project eventual performance [7, 8]. They operate on scalar loss or accuracy values and require multiple epochs to fit a reliable curve. Critically, they do not examine the internal structure of model behavior across classes and do not use any measure of how organized the model’s predictions have become.

Zero-cost proxy methods compute saliency scores, gradient norms, or related statistics from untrained or minimally trained weights [9, 10]. These methods operate in parameter space rather than behavioral space. They do not measure class-conditional behavior or quantify how consistently a model has begun recognizing distinct output categories, so they cannot directly assess whether a network is in the process of forming coherent task identity.

This paper takes a different approach. Rather than extrapolating a performance trajectory or scoring architecture parameters, we observe whether the architecture has begun organizing its predictions into coherent class-conditional patterns after one training epoch, and use that observation as a binary formation prediction. The key finding is that this signal, summarized as a single scalar derived from the confusion matrix, separates architectures that will form task identity from those that will not, across the tested architecture families and datasets, at a threshold that did not require adjustment between experiments.

The method connects to prior work on confusion matrix analysis for behavioral assessment [14, 15] but differs in both the question it answers and the information it requires. Prior work in this line used correlation between confusion matrices at two time points to measure behavioral stability or predict training speed. The present method uses the diagonal strength of a single confusion matrix at one time point to predict whether the architecture will reach task competence at all within the tested training budget.

The contributions of this paper are as follows. We identify a threshold in the identity deficit metric at $I_c = 0.22$ that separates architectures forming task identity from those that do not in the tested setting. We validate a binary prediction rule across 22 architectures, 3 datasets, and 2 network types achieving 95% accuracy from one training epoch. We document a consistent 80% competence cutoff that produced more reliable cross-dataset results than earlier per-dataset cutoffs. We characterize a caution zone above the threshold containing slow learners for which the single-epoch signal is genuinely ambiguous. And we report the negative result from the earlier variable-cutoff methodology that motivated the final design.

To our knowledge, prior work does not provide a single-epoch confusion-matrix-based viability rule of this kind.

2 The Method

2.1 Identity Deficit

Let CM denote the confusion matrix produced by a network after one training epoch, evaluated on a held-out validation set. For a classification task with N classes, CM is an $N \times N$ array where element (i, j) contains the count of samples with true label i and predicted label j . The formation score F is the diagonal strength of this matrix:

$$F = \frac{\text{tr}(\text{CM})}{\sum_{i,j} \text{CM}_{ij}} \quad (1)$$

In standard single-label classification, F is numerically equal to overall classification accuracy, since the trace counts correct predictions and the sum counts all predictions. The notation F is used throughout to emphasize that this quantity is being interpreted as a measure of behavioral formation rather than performance per se: it captures how strongly the network has begun sorting inputs into distinct output categories after one epoch of training.

The identity deficit is:

$$I_{\text{deficit}} = 1 - \sqrt{F} \quad (2)$$

The square root transformation places the viability boundary in the region where the early-epoch behavioral signal actually lies. An architecture with $F = 0.95$ after epoch one has $I_{\text{deficit}} = 0.025$, while an architecture with $F = 0.60$ has $I_{\text{deficit}} = 0.226$. The linear complement $1 - F$ would place the threshold at $F = 0.78$, requiring 78% accuracy after one epoch to be classified as viable, which is implausibly demanding for an early-epoch signal. The square root places the threshold at $F = (1 - 0.22)^2 \approx 0.608$, roughly 61% accuracy, which is a more natural location for a one-epoch formation boundary and preserves separation near the observed viability boundary without requiring an implausibly high early-epoch accuracy threshold.

2.2 Viability Classification

Given an architecture’s epoch-one identity deficit, the primary prediction rule is binary:

- $I_{\text{deficit}} < 0.22$: classified as *likely to form task identity* within the tested training budget.
- $I_{\text{deficit}} \geq 0.22$: classified as *likely non-viable* within the tested training budget.

In a conservative deployment policy, architectures with $0.22 \leq I_{\text{deficit}} \lesssim 0.40$ may be treated as a *caution zone* rather than terminated immediately. This range is motivated by the single misclassified case in the reported experiments, a slow-learning 2-layer MNIST CNN with $I_{\text{deficit}} = 0.3702$ that eventually converged to 93.88% after 30 epochs, and by the corresponding caution-zone language in the patent specification. Under this policy, architectures below 0.22 continue training, architectures between 0.22 and approximately 0.40 receive additional early-epoch evaluation before a final termination decision, and architectures above 0.40 are treated as likely non-viable.

The threshold $I_c = 0.22$ was held fixed across the five reported validation experiments.

2.3 Formed Identity Criterion

We define formed task identity as reaching a final classification accuracy of at least 80% on 10-class problems. This cutoff was chosen because 80% accuracy represents an eightfold improvement over the 10% random baseline, indicating genuine task competence rather than marginal above-chance performance. It is achievable across all tested datasets and architectures that the method classifies as viable, and it does not shift with dataset difficulty: the same 80% criterion applies to MNIST [1], Fashion-MNIST [2], and CIFAR-10 [3], even though the absolute performance ceilings of the tested architecture families differ substantially across those three benchmarks.

An earlier version of this work used per-dataset accuracy cutoffs (0.90 for MNIST, 0.85 for Fashion-MNIST, 0.65 for CIFAR-10), which produced a total prediction accuracy of 19/22 (86%) and only 1/4 (25%) on CIFAR-10. The lower CIFAR-10 cutoff allowed architectures converging between 58% and 79% to be labeled as having formed task identity, which is inconsistent with meaningful task competence on a 10-class problem. Adopting the consistent 80% cutoff improved total accuracy to 21/22 (95%) and CIFAR-10 accuracy to 4/4 (100%), because all four tested simple CNNs on CIFAR-10 fail to reach 80%, making them non-viable for this architecture family on this task within the tested budget.

3 Experiments

3.1 Datasets

All experiments used real image classification data loaded from `keras.datasets` with no synthetic augmentation. MNIST [1] consists of 60,000 training images and 10,000 test images of 28×28 grayscale handwritten digits across 10 classes. Fashion-MNIST [2] uses the same image dimensions and class count but replaces digits with clothing items, producing a harder classification problem with a lower performance ceiling for simple architectures. CIFAR-10 [3] consists of 50,000 training images and 10,000 test images of 32×32 RGB photographs across 10 classes. For all datasets, 10% of the training set was held out as a validation set for epoch-one confusion matrix construction. Test set accuracy is reported as the final performance figure.

3.2 Architectures

Two architecture families were evaluated. Multilayer perceptrons (MLPs) use fully connected layers with ReLU activations and no convolutional components. Convolutional neural networks (CNNs) use sequences of convolutional and pooling layers followed by a classification head.

The tiny MLP set consisted of five single-hidden-layer networks with 2, 3, 4, 5, and 8 neurons respectively, designed to span the viability boundary by stepping through minimal-capacity networks. The standard MLP set consisted of five networks: a single-layer model with 16 neurons, a single-layer model with 64 neurons, a 2-layer model with 128 neurons per layer, a 3-layer model with 256 neurons per layer, and a 5-layer model with 128 neurons per layer. The CNN sets consisted of four architectures each, ranging from 2 to 5 convolutional layers, tested independently on MNIST, Fashion-MNIST, and CIFAR-10.

3.3 Procedure

For each architecture, the procedure followed six steps. First, the network was initialized with random weights and trained for exactly one epoch using standard gradient-based optimization and backpropagation. Second, the trained network generated predictions on the held-out validation set. Third, a confusion matrix was constructed from those predictions and the ground-truth labels. Fourth, the formation score F and identity deficit I_{deficit} were computed. Fifth, a viability prediction was recorded: WILL FORM if $I_{\text{deficit}} < 0.22$, WILL FAIL otherwise. Sixth, the network was trained to the full reported budget (30 total epochs for MNIST and Fashion-MNIST, 50 total epochs for CIFAR-10) and the final test accuracy was compared against the 80% formed-identity cutoff to establish the ground-truth outcome.

The experiments were implemented in Keras/TensorFlow, with confusion matrices computed using

scikit-learn. Random seeds were set before each experiment. No synthetic data was used at any stage. Results were saved to timestamped JSON files at experiment completion.

4 Results

4.1 Baseline Validation: Bidirectional Threshold Sensitivity

The tiny MLP experiment was designed to test the threshold in both directions on the same dataset by stepping through architectures of increasing capacity from clearly non-viable to clearly viable. All five architectures were correctly classified. Table 1 reports the results.

Table 1: Tiny MLP validation on MNIST. Both failure and success cases are correctly predicted. The transition from Tiny-3 to Tiny-4 crosses the 0.22 threshold and the 80% formed-identity cutoff simultaneously.

Architecture	F	I_{deficit}	Prediction	Final Acc.	Correct
Tiny-2 (2 neurons)	0.395	0.371	WILL FAIL	65.49%	Yes
Tiny-3 (3 neurons)	0.600	0.225	WILL FAIL	77.77%	Yes
Tiny-4 (4 neurons)	0.753	0.132	WILL FORM	86.54%	Yes
Tiny-5 (5 neurons)	0.781	0.117	WILL FORM	89.21%	Yes
Tiny-8 (8 neurons)	0.864	0.070	WILL FORM	92.92%	Yes

The transition from Tiny-3 to Tiny-4 is the sharpest boundary in this experiment. Adding one neuron reduces I_{deficit} by 0.093 and raises final accuracy by 8.77 percentage points, with both the threshold and the 80% competence cutoff crossed in the same step. This is consistent with a phase transition interpretation in which one unit of capacity moves the architecture across a regime boundary rather than producing a smooth incremental improvement.

4.2 Standard MLP Validation

Five standard MLP architectures were tested on MNIST. All produced I_{deficit} substantially below the threshold after one epoch and all correctly reached formed identity. Table 2 reports the results.

Table 2: Standard MLP validation on MNIST. All five architectures are correctly classified as viable. Identity deficit values range from 0.020 to 0.047, well below the 0.22 threshold.

Architecture	F	I_{deficit}	Prediction	Final Acc.	Correct
Tiny MLP (16)	0.908	0.047	WILL FORM	95.38%	Yes
Small MLP (64)	0.937	0.032	WILL FORM	97.07%	Yes
Medium MLP (128x2)	0.953	0.024	WILL FORM	97.86%	Yes
Large MLP (256x3)	0.961	0.020	WILL FORM	98.16%	Yes
Deep MLP (128x5)	0.954	0.023	WILL FORM	97.94%	Yes

All five architectures formed task identity rapidly on MNIST, with final accuracies clustering between 95.38% and 98.16%. The I_{deficit} values in this experiment are an order of magnitude below the threshold, confirming that viable MLP architectures on a well-suited dataset produce a strong and unambiguous epoch-one signal.

4.3 CNN Validation on MNIST

Four CNN architectures of increasing depth were tested on MNIST. This experiment produced the method’s single misclassification. Table 3 reports the results.

Table 3: CNN validation on MNIST. The 2-layer CNN is the method’s single false negative: predicted non-viable but converged to 93.88% after 30 epochs. Its identity deficit of 0.370 places it in the caution zone. The 3-layer through 5-layer CNNs are all correctly classified.

Architecture	F	I_{deficit}	Prediction	Final Acc.	Correct
2-layer CNN	0.397	0.370	WILL FAIL	93.88%	No
3-layer CNN	0.791	0.110	WILL FORM	98.24%	Yes
4-layer CNN	0.936	0.033	WILL FORM	99.08%	Yes
5-layer CNN	0.968	0.016	WILL FORM	99.39%	Yes

The 2-layer CNN had $I_{\text{deficit}} = 0.370$ after one epoch and was predicted non-viable. It eventually converged to 93.88% after 30 total epochs of training, making it a false negative. Its identity deficit falls in the caution zone (0.22 to approximately 0.40), and the method treats architectures in this range as candidates for additional early-epoch evaluation rather than immediate termination.

The gap between the 2-layer and 3-layer CNNs is the largest architecture-to-architecture drop in the dataset. I_{deficit} falls by 0.260 and final accuracy rises by 4.36 percentage points from one additional convolutional layer. The 4-layer and 5-layer CNNs continue to much lower I_{deficit} values (0.033 and 0.016), with final accuracies above 99%.

4.4 CNN Validation on Fashion-MNIST

Four CNN architectures were tested on Fashion-MNIST. All four were correctly classified as viable. Table 4 reports the results.

Table 4: CNN validation on Fashion-MNIST. All four architectures are correctly classified as viable. The 2-layer CNN has $I_{\text{deficit}} = 0.219$, just below the 0.22 threshold, and converges to 83.41%, just above the 80% competence cutoff.

Architecture	F	I_{deficit}	Prediction	Final Acc.	Correct
2-layer CNN	0.610	0.219	WILL FORM	83.41%	Yes
3-layer CNN	0.636	0.203	WILL FORM	89.63%	Yes
4-layer CNN	0.773	0.121	WILL FORM	91.40%	Yes
5-layer CNN	0.782	0.116	WILL FORM	91.58%	Yes

The 2-layer CNN result is the closest near-threshold success case in the dataset. At $I_{\text{deficit}} = 0.219$, it sits 0.003 below the threshold and converges to 83.41%, 3.41 percentage points above the competence cutoff. The method correctly predicts it as viable. This near-threshold success is useful because it shows the method can retain borderline viable architectures rather than treating every low-capacity CNN as non-viable. The same nominal 2-layer depth does not produce the same epoch-one formation signal across datasets, indicating that the early formation signal is determined by the interaction between architecture and task, not by depth alone.

4.5 CNN Validation on CIFAR-10

Four CNN architectures were tested on CIFAR-10 with a training budget of 50 total epochs. All four were correctly classified as non-viable. Table 5 reports the results.

Table 5: CNN validation on CIFAR-10. All four architectures are correctly classified as non-viable. None reaches the 80% formed-identity cutoff after 50 epochs. The 5-layer CNN comes closest at 79.53%.

Architecture	F	I_{deficit}	Prediction	Final Acc.	Correct
2-layer CNN	0.308	0.445	WILL FAIL	58.65%	Yes
3-layer CNN	0.371	0.391	WILL FAIL	74.97%	Yes
4-layer CNN	0.407	0.362	WILL FAIL	77.77%	Yes
5-layer CNN	0.433	0.342	WILL FAIL	79.53%	Yes

None of the four tested CNN architectures reaches 80% on CIFAR-10 within 50 training epochs. The 5-layer CNN comes closest at 79.53%, 0.47 percentage points below the competence cutoff. Three of the four I_{deficit} values fall in the caution zone (0.342 to 0.391) and one falls above it (0.445). All four predictions are correct. This experiment demonstrates that the method correctly identifies a capacity-task mismatch: simple convolutional architectures of 2 to 5 layers are not viable on 10-class RGB image classification within this training budget, and their epoch-one formation scores reflect this before any extended training is conducted.

The CIFAR-10 result also illustrates why a consistent 80% competence cutoff is necessary. Under the earlier per-dataset cutoff of 65% for CIFAR-10, three of the four architectures would have been labeled as having formed identity despite convergence accuracies of 74.97%, 77.77%, and 79.53%, producing three false negatives and a CIFAR-10 accuracy of 1/4. The uniform 80% criterion avoids that inconsistency.

4.6 Summary

Table 6 summarizes all five experiments.

Table 6: Summary of all validation experiments. Total accuracy is 21/22 (95%) across 22 architectures, 3 datasets, and 2 network types. The single false negative is the MNIST 2-layer CNN, a slow learner in the caution zone.

Test	Dataset	Network	Result	Accuracy
1	MNIST	Tiny MLP	5/5	100%
2	MNIST	MLP	5/5	100%
3	MNIST	CNN	3/4	75%
4	Fashion-MNIST	CNN	4/4	100%
5	CIFAR-10	CNN	4/4	100%
Total			21/22	95%

Across all 22 tested cases, the method produced zero false positives. No architecture predicted as likely to form task identity failed to reach the 80% competence cutoff. The single false negative is the MNIST 2-layer CNN with $I_{\text{deficit}} = 0.370$, which sits in the caution zone and represents a slow learner rather than a genuinely non-viable architecture. The caution zone partition of all 22

cases is as follows: 15 cases fell below 0.22 with 15 correct predictions and zero errors, 6 cases fell in the caution zone (0.22 to approximately 0.40) with 5 correct predictions and 1 false negative, and 1 case fell above 0.40 with 1 correct prediction. The asymmetry is operationally favorable: the method never silently passed a failing architecture.

5 Discussion

5.1 What the Threshold Captures

The central finding is that a single scalar computed from one confusion matrix after one training epoch carries enough information to predict whether an architecture will form task identity within the tested training budgets, at a threshold that did not require adjustment across the tested settings. The natural question is what that scalar is actually measuring.

In the trace/sum setting used here, the formation score F is numerically equal to classification accuracy at epoch one. The identity deficit $I_{\text{deficit}} = 1 - \sqrt{F}$ is a nonlinear transformation of that accuracy that expands the scale near the viability boundary. An architecture with I_{deficit} well below 0.22 has, after one pass through the training data, already organized its predictions into patterns that largely match the target classes. An architecture with I_{deficit} well above 0.22 has not, and the question the method addresses is whether that gap will close given additional training.

In the tested set, architectures below the implied formation boundary at epoch one usually did not reach 80% within the tested training budgets, with one slow-learning exception in the caution zone. This is not a trivial observation because epoch one is a long way from convergence for most architectures, and the initial organization of predictions reflects something about the architecture’s capacity and the difficulty of the task that persists through training.

5.2 The Phase Transition Interpretation

The tiny MLP experiment is the clearest evidence for a phase transition reading of the threshold. The transition from Tiny-3 to Tiny-4, a change of one neuron, crosses the identity deficit threshold and the competence cutoff simultaneously. Tiny-3 at 3 neurons produces $I_{\text{deficit}} = 0.225$ and converges to 77.77%. Tiny-4 at 4 neurons produces $I_{\text{deficit}} = 0.132$ and converges to 86.54%. The gap is discontinuous relative to the size of the architectural change: adding one neuron drops I_{deficit} by 0.093 and raises final accuracy by nearly 9 percentage points.

The MNIST CNN family shows a similar pattern. The 2-layer CNN is in the caution zone at $I_{\text{deficit}} = 0.370$ while the 3-layer CNN sits at 0.110, well below the threshold. Both use the same training procedure, and the architectural difference is one layer. The fact that one layer makes such a large difference to the epoch-one formation signal is consistent with the idea that there is a minimum depth requirement for organizing MNIST features into coherent class-conditional representations during the first training pass.

The phase-transition interpretation used here is meant in this empirical sense: the data shows a boundary rather than a smooth continuum, and the threshold marks that boundary in the tested cases. It is not a claim that this is a formal phase transition in the statistical physics sense.

5.3 The Caution Zone and Asymmetric Error Structure

The validation results show a favorable asymmetry. Zero false positives were observed across all 22 tested cases, while one false negative appeared in the caution zone. This means the method, in the tested setting, never approved a failing architecture. It flagged one eventually viable architecture as potentially non-viable, but that architecture had $I_{\text{deficit}} = 0.370$, which the patent specification and this paper both treat as a region warranting additional evaluation rather than immediate termination.

The operational implication is that, in the tested setting, false negatives appeared only in the 0.22 to 0.40 range, while zero false positives were observed overall. For NAS applications where wasted training on failing architectures is more costly than extending evaluation for a small caution-zone subset, this asymmetry is desirable. For applications where missing a viable architecture is the bigger concern, the caution zone policy provides a path to recover slow learners before terminating them.

5.4 Connection to Prior Work in This Line

This method is closely related to earlier confusion-matrix-based behavioral metrics developed in prior work. The task-identity method [14] uses Pearson correlation between flattened confusion matrices at two time points to detect behavioral drift in deployed models. The formation detection method [15] uses correlation between a one-epoch confusion matrix and a mature checkpoint to predict how much additional training a viable architecture will require. The present method uses the diagonal strength of a single one-epoch confusion matrix to predict whether the architecture will reach task competence at all.

The three methods ask different questions from the same type of input. Task-identity asks whether behavior has changed between two observations. Formation asks how fast a known-viable architecture will train. The present method asks whether an architecture is viable at all within a practical budget. The inputs differ accordingly: task-identity requires two model snapshots and a labeled evaluation set, formation requires one early snapshot and one mature snapshot, and the present method requires only the one-epoch snapshot.

6 Limitations

The validation in this paper is bounded to 10-class image classification on three benchmarks using MLP and simple CNN families. Several limitations follow directly from this scope.

The threshold $I_c = 0.22$ and the 80% competence cutoff were validated only on 10-class problems. Whether the present threshold transfers to classification problems with substantially different class counts has not been tested.

The architecture families tested are restricted to MLPs and simple CNNs with up to five layers. Whether the threshold holds for residual networks, attention-based architectures, or other modern families is not established by this validation. The method makes predictions about what epoch-one behavior implies for eventual convergence under the specific training budgets used here. A longer training budget might change the ground-truth labels for some caution-zone architectures.

The caution zone boundary at approximately 0.40 is an observation from the patent specification and the single misclassified case rather than a separately derived threshold. Whether 0.40 is the

right upper boundary for the caution zone on other architecture families or datasets has not been tested.

The results are also dependent on the 80% competence cutoff. The 5-layer CIFAR-10 CNN converged at 79.53%, 0.47 percentage points below the cutoff. The method’s predictions are correct with respect to the 80% criterion but should not be read as claims about viability at other competence levels.

Finally, the validation set for each experiment is 10% of the training data, and the epoch-one confusion matrix is constructed from that validation set. The identity deficit estimate at epoch one has variance that depends on validation set size, class balance, and random initialization. Averaging I_{deficit} across multiple random seeds or across the first few epochs may produce a more stable estimate in practice, particularly for architectures near the threshold.

7 Related Work

7.1 Early Stopping and Resource Schedulers

Early stopping with patience-based criteria is the standard baseline for terminating underperforming training runs [4]. Successive Halving [6] and Hyperband [5] extend this to multi-fidelity settings, allocating progressively more resources to configurations that survive early evaluation rounds. These methods are effective at filtering configurations that are slow to improve but are reactive by design: they require observing a trend across multiple epochs before any termination decision is possible. The present method makes its prediction from a single epoch without requiring a trend.

7.2 Learning Curve Extrapolation

Domhan et al. [7] proposed fitting parametric learning curves to early training data to project final performance. Baker et al. [8] incorporated learning curve features into a reinforcement learning based NAS framework. These methods rely on scalar performance trajectories and require multiple training epochs to fit a curve. They do not use the structure of the model’s predictions across classes and do not produce a formation-based viability signal from a single snapshot.

7.3 Zero-Cost Proxies

A line of work attempts to score architectures before or after minimal training using parameter-space statistics. Mellor et al. [9] proposed a proxy based on the correlation structure of activations at initialization. Abdelfattah et al. [10] evaluated a range of zero-cost proxies including gradient norms, activation statistics, and weight norms. These methods operate without labeled target data and require very little computation. The present method differs in that it operates in behavioral space after one epoch of training on labeled data, using the class-conditional structure of predictions rather than parameter-space statistics.

7.4 Phase Transitions in Neural Network Training

The double-descent phenomenon [11, 12] describes non-monotonic generalization behavior as a function of model size or training time. The grokking phenomenon [13] describes delayed generalization that occurs long after training loss has converged. Both lines of work characterize qualitative transitions in network behavior but are descriptive and post-hoc: they explain what happened after

training rather than providing an early-epoch signal that predicts whether a transition will occur. The present method is predictive rather than descriptive.

7.5 Confusion Matrix Methods for Behavioral Assessment

Prior work in this line used confusion matrix correlation to measure behavioral stability and predict training efficiency. The task-identity method [14] showed that Pearson correlation between flattened confusion matrices at two time points detects behavioral drift that accuracy-based metrics miss, including cases where accuracy is stable but the pattern of errors has shifted substantially. The formation detection method [15] showed that the correlation between an epoch-one confusion matrix and a mature checkpoint predicts total training time needed across MLP architectures on easy and medium-difficulty benchmarks. The present method extends this line by using the diagonal strength of a single confusion matrix rather than a correlation between two matrices, and by addressing viability rather than speed or stability [16].

8 Conclusion

We presented a method for predicting whether a neural network architecture will form task identity within a practical training budget using one confusion matrix constructed after one training epoch. The identity deficit $I_{\text{deficit}} = 1 - \sqrt{F}$, where F is confusion matrix diagonal strength, separates viable from non-viable architectures at a threshold of $I_c = 0.22$ across the tested settings.

Across 22 architectures spanning MLPs and CNNs on MNIST, Fashion-MNIST, and CIFAR-10, the method correctly classified 21 of 22 cases using a consistent 80% competence cutoff. The single misclassification is a slow learner whose identity deficit falls in the caution zone (0.22 to approximately 0.40) that the method treats as a region for additional evaluation rather than immediate termination. Zero false positives were observed in the tested setting.

The threshold was held fixed across the five validation experiments without per-dataset or per-architecture adjustment. The method requires no access to gradients, weights, or loss curves, and produces its prediction from one forward pass through a validation set after one training epoch.

The results are bounded to the tested architecture families and benchmarks. Extension to other architecture types, task formulations, or training regimes requires additional validation.

Code Availability

All validation code, results, and timestamped JSON outputs are available at <https://github.com/Wise314/neural-phase-transition-detection>.

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Patent Disclosures

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